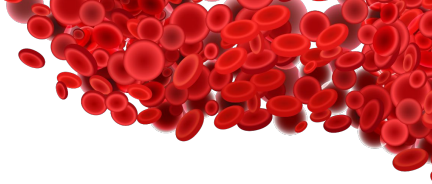
食物配方對凝血時間之影響

第九組:統計115 H24116065 黃柏翰
統計115 H24116112 張智賢



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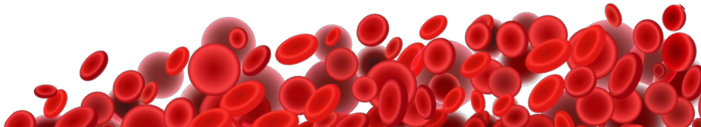
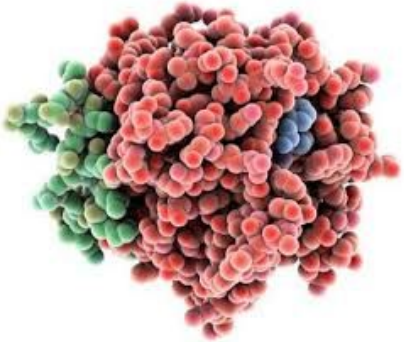
資料介紹



凝血機制與飲食的關聯

- 血液凝固
  - 在血管受損時，將液態血液變為固態血塊
  - 主要目的：止血、保護傷口
  - 血小板：最先抵達破損處的先鋒部隊
  - 凝血因子：負責建造穩固結構的蛋白質

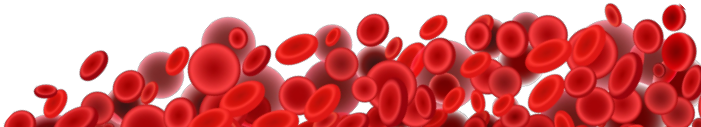
- 飲食如何影響凝血
  - 身體無法自行製造凝血因子
  - 其生產原料完全依賴食物中的營養素



- 資料來源
  - 《Statistics for Experimenters》(G.E.P. Box, W.G. Hunter, J.S. Hunter, 1978)
- 實驗設計(CRD)
  - 對象：24 隻實驗動物，隨機分為 4 組。
  - 自變項 (控制變因)：食物 (diet)，共 A, B, C, D 四種不同配方。
  - 應變項 (測量結果)：凝血時間 (coagulation time)，單位為秒。
- 研究目的
  - 探討不同的食物配方對動物的血液凝固時間是否造成影響

02

統計分析



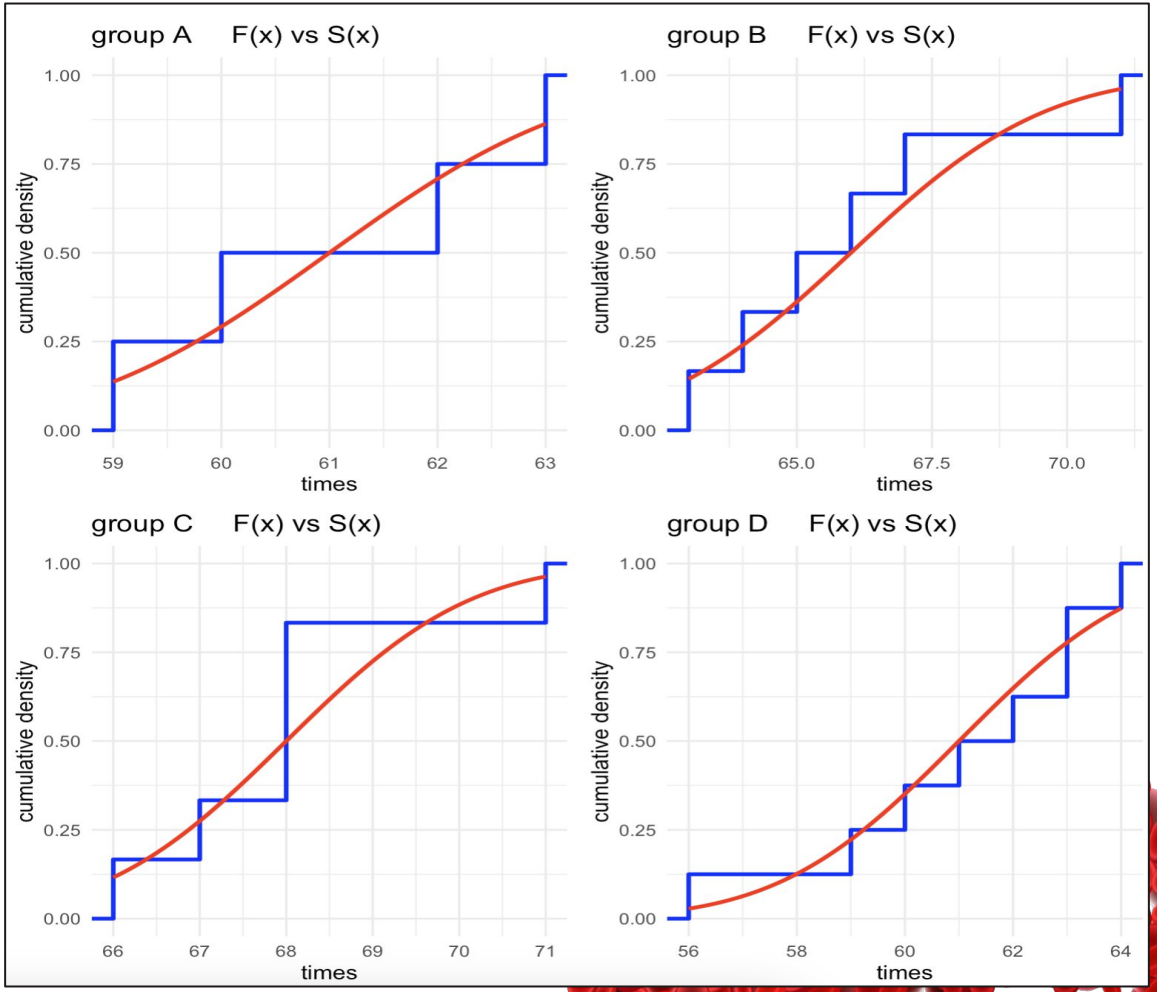
1. Kolmogorov–Smirnov test

H_0：數據服從常態分配
H_a：數據不服從常態分配

$$T = \max|F(x) - S(x)|$$

$$RR = \{T \geq KS_{\alpha/n}(n)\}$$

	Group	T	P-value	Decision
1	A	0.2081	0.9817	不拒絕虛無假設
2	B	0.1952	0.9427	
3	C	0.3333	0.5176	
4	D	0.1525	0.9923	


group A F(x) vs S(x)
group B F(x) vs S(x)
group C F(x) vs S(x)
group D F(x) vs S(x)
cumulative density
times
1.00
0.75
0.50
0.25
0.00
59
60
61
62
63
65.0
67.5
70.0
66
67
68
69
70
71
56
58
60
62
64

2. Levene's Test

H_0 ：$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$

H_a ： $\sigma_1^2, \sigma_2^2, \sigma_3^2, \sigma_4^2$ 不完全相等

$D_{ij} = |Y_{ij} - \eta_i|$, $\mathrm{i} = 1,2,3,4$ $\mathrm{j} = 1,2,\ldots,n_i$

	Df	F value	Pr (>F)
Treatment	3	0.6492	0.5926
Error	20		

P-value = 0.5926 > 0.05

故不拒絕虛無假設

3. ANOVA

H_0 ：$\mu_1 = \mu_2 = \mu_3 = \mu_4$

H_a ：$\mu_1, \mu_2, \mu_3, \mu_4$不完全相等

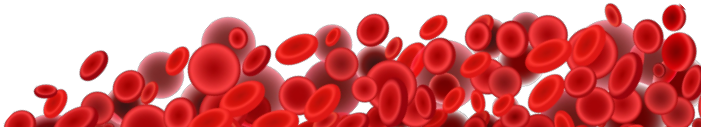
	df	SS	MS	F-value	Pr(>F)
Treatment	3	228	76	13.571	4.658e-05
Error	20	112	5.6		

P-value = 4.658e-05 < 0.05

故拒絕虛無假設

03

事後檢定



Tukey-Kramer procedure

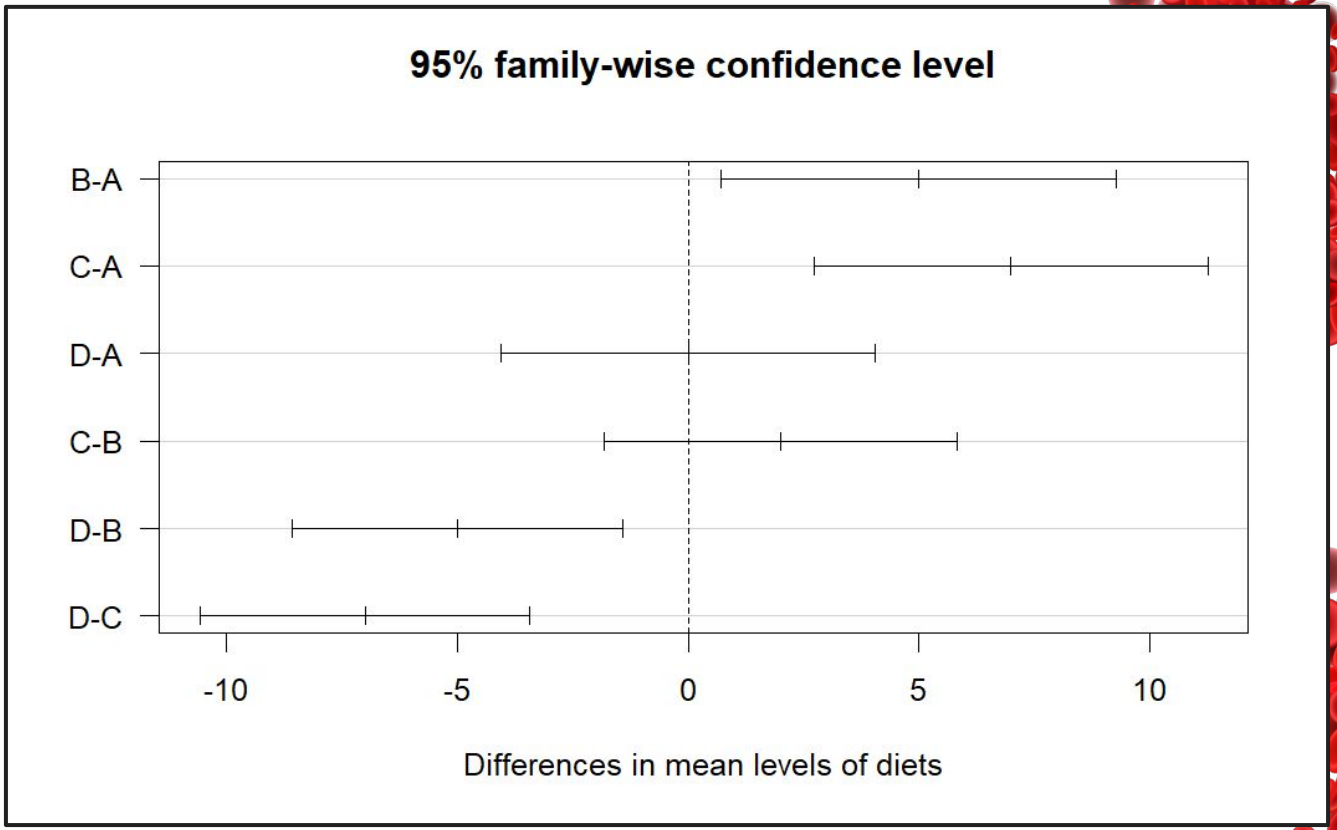
$$H_0 : \mu_i = \mu_j$$
$$H_a : \mu_i \neq \mu_j$$

$\mu_i - \mu_j$ 的$(1-\alpha)100\%\ C.I.$:

$$\bar{X}_i - \bar{X}_j \pm \frac{q_{1-\alpha, df=(N-k)}}{\sqrt{2}} \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

$$reject\ H_0, if\ 0 \notin [L, U]$$

$$do\ not\ reject\ H_0, if\ 0 \in [L, U]$$

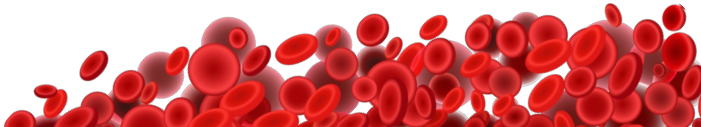


$$reject\ H_0 : \mu_A = \mu_B\ , H_0 : \mu_A = \mu_C\ ,$$

$$H_0 : \mu_D = \mu_C\ , H_0 : \mu_B = \mu_D$$

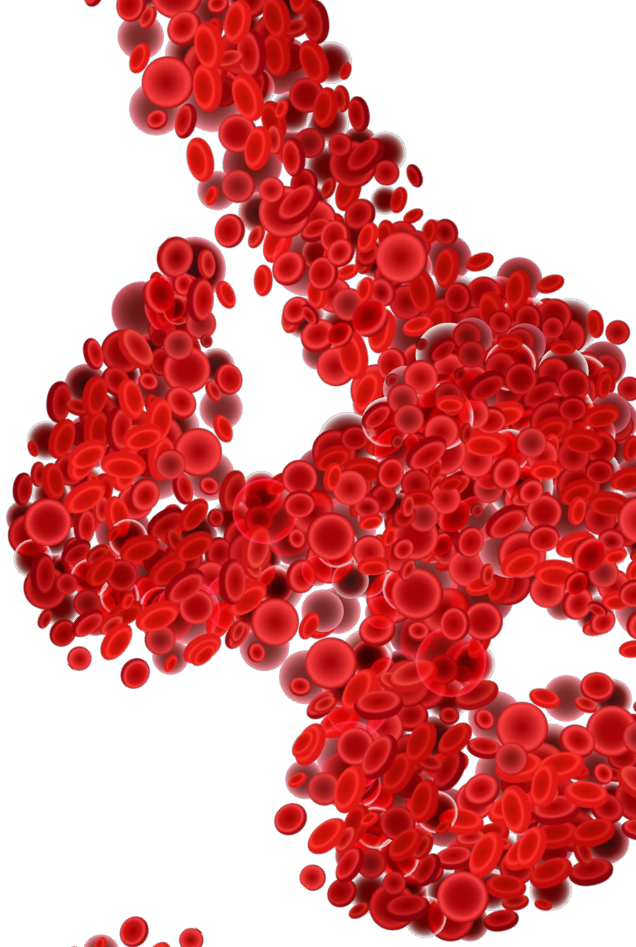
04

結論



結論：不同飲食確實會影響凝固時間

- A-D、C-B之間效果類似，凝固時間沒有顯著差異。
- 其他組：有顯著差異，不能算相同。

References

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